MIL-A-14822(MU)
21 October 1968

SUPERSEDING
FA(FC) PD-72
8 May 1961

MILITARY SPECIFICATION

AMPLIFIER - DETECTOR: 10525674

1. SCOPE

1.1 This specification covers an I. F. Amplifier-Detector used in Radar Chronograph M36. The I. F. Amplifier-Detector, hereinafter referred to as the I. F. Amplifier, is used to amplify the receiver I. F. signal from a microwave mixer and to detect the doppler-frequency modulation.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction, M18 Manufacturing Process and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

FSC 1290

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DRAWING

U. S. Army, Frankford Arsenal

D10525674 Amplifier-Detector

PACKAGING DATA SHEET

10525674 Packaging of Amplifier-Detector

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The I. F. Amplifier shall be manufactured in accordance with Drawing D10525674 and drawings pertaining thereto.

3.2 General Specifications.- The I. F. Amplifier shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.2.1 MIL-F-13926

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Workmanship
(e) Soldering electrical connections

3.2.2 MIL-C-46357

(a) Wiring
(b) Thread fasteners
(c) Marking
(d) Finishes
(e) Etched circuit assemblies

3.3 ENVIRONMENTAL

3.3.1 Shock.- The I. F. Amplifier shall be capable of three (3) shocks of 50 gravity units (g's) of eleven (11) millisecond duration in each of the following directions:

(a) Bottom down
(b) Front down
(c) Right side down
(d) Back down
(e) Left side down
(f) Top down

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Subsequent to the above shock, the I. F. Amplifier shall not have been damaged or adversely affected, and shall meet the requirements of 3.5

3.3.2 Vibration.- The I. F. Amplifier shall be capable of continuous operation while being vibrated in a simple harmonic motion having an amplitude of 0.03 (0.06 inch maximum total excursion) with the frequency being varied uniformly between the limits of ten (10) and fifty-five (55) hertz. The entire frequency range, from ten (10) to fifty-five (55) hertz and return to ten (10) hertz, shall be uniformly traversed in one (1) minute plus or minus 15 seconds. This motion shall be applied for a period of thirty (30) minutes plus or minus one (1) minute in each of three (3) mutually perpendicular directions (x, y, z axis) for a total of 1.5 hours. During the above vibration, the I. F. Amplifier shall meet the requirements of 3.5 with the exception of 3.5.3 once in each plane At all other times, the I. F. Amplifier shall have power applied. Subsequent to the above vibration, the I. F. Amplifier shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.3.3 Humidity.- The I. F. Amplifier shall be capable of continuous operation during four (4) hours plus or minus ten (10) minutes of continuous exposure to a relative humidity of ninety (90) to ninety-five (95) per cent at a temperature of plus 100°F plus or minus 5°F. During and su bsequent to the above humidity, the I. F. Amplifier shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.3.4 Sand and Dust.- The I. F. Amplifier shall be capable of withstanding three (3) hours of continuous exposure to blowing sand of particle size 0.01 to 1.00 mm and blowing dust of particle size 0.0001 to 0.01 mm with a density between 0.1 to 0.5 gram per cubic foot of dust at a temperature of plus 75°F plus or minus 5°F, a relative humidity between fifteen (15) to thirty (30) percent, and a wind velocity of fifteen (15) knots. Subsequent to the above sand and dust exposure, the I. F. Amplifier shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.3.5 Altitude.

3.3.5.1 Operating.- The I. F. Amplifier shall be capable of continuous operation during four (4) hours of continuous exposure to an altitude of 10,000 feet at a temperature of plus 75°F. During and subsequent to the above exposure, the I. F. Amplifier shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.3.5.2 Non-Operating.- The I. F. Amplifier shall be capable of withstanding four (4) hours of continuous exposure to an altitude of 50,000 feet at a temperature of minus 65°F. Subsequent to the above exposure, the I. F. Amplifier shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

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3.3.6 Fungus.- The I.F. Amplifier shall be capable of withstanding twenty-eight (28) days exposure to a composite spore suspension consisting of the following cultures:

(a) Chaetomuim globsom
(b) Aspergillus flavus
(c) Aspergillis versicalor
(d) Penicillum luteum
(e) Rhizopus nigricans

The composite spore suspension shall be incubated at an ambient temperature of 80°F plus or minus 4°F with an RH of 87 plus or minus 2%. Subsequent to the above exposure, the I. F. Amplifier shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.3.7 Storage Temperatures.- The I. F. Amplifier shall be capable of withstanding temperature stabilization at plus 160°F and minus 80°F for a period of eight (8) hours at each temperature extreme. Subsequent to each of the above temperature extremes (+160°F and -80°F) after temperature stabilization at plus 70°F plus or minus 15°F, the I. F. Amplifier shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.3.8 Operating Temperatures.- The I. F. Amplifier shall be capable of continuous operation for eight (8) hours at plus 140°F. The I. F. Amplifier shall also be capable of continuous operation for eight (8) hours at minus 50°F. During and subsequent to the above temperature extremes (+140°F and -50°F), the I. F. Amplifier shall not have been damaged or adversely affected and shall meet the requirements of 3.5 at least once at the beginning, middle, and ending of each eight (8) hour temperature extreme. There shall be a five (5) to six (6) minute warm-up time prior to performing the first test of 3.5 at -50°F.

3.4 Electrical.

3.4.1 Continuity.- The I. F. Amplifier shall demonstrate continuity in accordance with the electrical schematic, Drawing F 10525694.

3.5 Performance.- The performance requirements specified herein shall be met on every I. F. Amplifier at Standard Ambient Conditions prior to government acceptance.

3.5.1 Power Supply.- All requirements of 3.5 shall be met over the following power supply ranges except where a paragraph specifically indicates an exception:

(a) +21.5 to +22.5 VDC
(b) +130 to +140 VDC
(c) 6.3 to 6.9 VAC

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3.5.2 Local Oscillator Frequency.- The local oscillator frequency shall be 45 megahertz plus or minus 0.01 percent.

3.5.3 Modulator Drive.- The modulator drive shall be capable of being adjusted between 0.2 and 0.5 volts RMS with a frequency of 45 megahertz plus or minus 0.01 percent into a 50 ohm load.

3.5.4 Modulator Drive Level.- The modulator drive le vel shall be adjusted to 0.4 volts RMS with a frequency of 45 megahertz plus or minus 0.01 percent into a 50 ohm load. Paragraph 3.5.1(a) shall be held at +22.0VDC.

3.5.5 Bandpass and I. F. Alignment.- The bandpass shall be between 0.26 and 1.0 megahertz with a center frequency of 45 megahertz plus or minus 0.01 percent and shall be adjusted for maximum amplitude at a modulation frequency of 120 kilohertz.

3.5.6 Gain.- Modulation Frequency Range of 10 to 120 kilohertz.

3.5.6.1 Signal Applied to J3.- With a 100 micro-volt (RMS) input to J3, the output signal into a 2500 ohm resistive load shall be an undistorted sine wave between 65 and 100 millivolts peak-to-peak over the modulation frequency range of 10 to 75 kilohertz and shall be an undistorted sine wave between 55 and 100 millivolts peak-to-peak over the modulation frequency range of 75 to 120 kilohertz.

3.3.6.2 Signal Applied to J4.- With a 100 micro-volt (RMS) input to J4, the output signal into a 2500 ohm resistive load shall be an undistorted sine wave between 65 and 100 millivolts peak-to-peak over the modulation frequency range of 10 to 75 kilohertz and shall be an undistorted sine wave between 55 and 100 millivolts peak-to-peak over the modulation frequency range of 75 to 120 kilohertz.

3.5.7 Noise and Distortion.- With no signal input, the output noise into a 2500 ohm resistive load shall be less than 0.8 millivolts peak-to-peak. With a 5 to 500 microvolt (RMS) input signal, the output shall be an undistorted sine wave over the modulating frequency range of 10 to 120 kilohertz.

3.5.8 Leakage.- The d.c. voltage across an 1040 ohm resistive load with no signal input shall be a level between +67 and +73 millivolts.

3.5.9 AFC DISC Meter.- With a +195 millivolt input level, the Monitor Meter shall read between +40 and +55 micro-amperes with the Monitor Selector Switch in the AFC DISC position. With a -195 millivolt input level, the Monitor Meter shall read between -40 and -55 micro-amperes with the Monitor Selector Switch in the AFC DISC position.

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3.5.10 MOD DRIVE Meter.- The Monitor Meter shall read between +40 and +70 micro-amperes with the Monitor Selector Switch in the MOD DRIVE position after the requirements of paragraph 3.5.4 have been accomplished.

3.5.11 MOD XTAL Meter.- With a signal output of 0.4 volts RMS at a frequency of 45 megahertz, the Monitor Meter shall read between -50 and -100 micro-amperes with an IN263 diode across the output having its anode grounded and shall read between +50 and +100 micro-amperes with the IN263 diode across the output having its cathode grounded for both MOD XTAL 1 and MOD XTAL 2 position of the Monitor Selector Switch.

3.5.12 MXR XTAL Meter.- With a signal input of 320 millivolts (RMS) at 45 megahertz across a IN263 diode, the Monitor Meter shall read between -50 and -100 micro-amperes with the diode having its anode grounded and shall read between +50 and +100 micro-amperes with the diode having its cathode grounded for both MXR XTAL 1 and MXR XTAL 2 positions of the Monitor Selector Switch.

3.6 Reliability Run-in.- In order to maximize component reliability, prior to acceptance by the Government, each I. F. Amplifier shall be subject to the following run-in conditions:

3.6.1 Temperature Run-in.- The I. F. Amplifier, without power applied, shall be temperature stabilized at plus 125°F and minus 25°F. At each temperature extreme, after temperature stabilization, power shall be applied and each unit shall be tested to all the performance requirements of 3.5 for a period of three (3) hours with a minimum of one (1) complete test to the requirements of 3.5 occurring at the beginning, middle and end of the three (3) hour period. There shall be a five (5) to six (6) minute warm-up time prior to the first performance test at -50°F. Subsequent to the above tests, the I. F. Amplifier shall not have been damaged or adversely affected, and shall meet the requirements of 3.5.

3.6.2 Vibration Run-in.- The I. F. Amplifier shall be capable of continuous operation while being vibrated in a simple harmonic motion having an amplitude of 0.03 (0.06 inch maximum total excursion) with the frequency being varied uniformly between the limits of ten (10) and fifty-five (55) hertz and return to ten (10) hertz and shall be uniformly traversed in one (1) minute. This motion shall be applied for a period of ten (10) minutes in each of three (3) mutually perpendicular directions (x, y, z axis) for a total of 0.5 hours. During the above vibration, the I. F. Amplifier shall meet the requirements of pars. 3.5.2, 3.5.4, 3.5.5, 3.5.6, and 3.5.7 once in each plane. Subsequent to the above vibration, the I. F. Amplifier shall not have been damaged or adversely affected, and shall meet the requirements of 3.5.

3.6.3 Shock Run-in.- The I. F. Amplifier shall be capable of three (3) shocks of 50 gravity units (g's) of eleven (11) millisecond duration in each

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of the following directions:

(a) Bottom down
(b) Front down
(c) Right side down
(d) Back down
(e) Left side down
(f) Top down

Subsequent to the above shock, the I. F. Amplifier shall not have been damaged or adversely affected, and shall meet the requirements of 3.5.

3.6.4 Failures During Reliability Run-in.- Replacement of defective components and correction of workmanship defects may be made during run-in under the following conditions:

(a) Any part, module, component or subassembly that is used as a replacement shall have previously passed environmental tests consistent with the reliability run-in environmental requirements of the I. F. Amplifier.

(b) Any replacement part shall have been exposed for at least one (1) hour at the extreme temperature being run immediately previous to being used as a replacement for any failure occurring during a temperature extreme.

(c) After a correction has been accomplished, at least one (1) hour of operation with a minimum of one (1) test to the requirements of 3.5 shall be successfully accomplished at the temperature at which the failure occurred.

(d) A failure during paragraphs 3.6.2 or 3.6.3 shall necessitate a complete re-run of the conditions of paragraphs 3.6.2 and 3.6.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General Provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First Article (Initial Production) Approval.- The requirements for first article approval and the responsibility (Government or Contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) I. F. Amplifiers manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

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4.2.1 Government Testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor Testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the government for approval. The government reserves the right to witness the contractor's inspection.

4.3 Inspection Provisions.

4.3.1 Examination and Test.

4.3.2 Acceptance and Rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special Sampling.- At the beginning of production and subsequent to First Article Approval (see 4.2), three units shall be selected by a Government representative and two from each fifty (50) produced. The sample shall have successfully met all other examinations and acceptance tests specified herein, and shall meet the following requirements tests:

	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
401.	Fabrication	3.1	Applic dwgs & Specs - Visual
402.	General Specifications	3.2	MIL-F-13926 & MIL-C-46357
403.	Shock	3.3.1	4.6.1.1
404.	Vibration	3.3.2	4.6.1.2
405.	Humidity	3.3.3	4.6.1.3
406.	Sand and Dust	3.3.4	4.6.1.4
407.	Altitude-operating	3.3.5.1	4.6.1.5
408.	Altitude-non-operating	3.3.5.2	4.6.1.5.1
409.	Storage temperatures	3.3.7	4.6.1.7.1
410.	Operating temperatures	3.3.8	4.6.1.7.2

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4.4.1 Failure of Sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor, and resubmitted samples have been approved (See 4.3.2).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government Furnished Inspection Equipment.- Where the contract provides for Government-Furnished Test Equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor Furnished Inspection Equipment

4.5.2.1 Contractor Design.- The contractor shall design and supply inspection equipment compatible with "Test Methods and Procedures" specified in 4.6 of this specification, and with the "Component Inspection" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test Methods and Procedures

4.6.1 Environmental.- Test equipment and test facilities utilized in the following environmental tests shall be in accordance with "Test Facilities" requirements of MIL-F-13926 and conditions specified in 3.3.

4.6.1.1 Shock.- The shock test shall be performed as specified in MIL-STD-810, Method 516, Procedure IV at the conditions specified in 3.3.1. Subsequent to the eighteen (18) shocks, the I. F. Amplifier shall be visually examined for Physical Damage, then tested as specified in 4.6.2.

4.6.1.2 Vibration.- The vibration test shall be performed as specified in MIL-STD-810, Method 514 at the conditions specified in 3.3.2. During the above vibration, the I. F. Amplifier shall meet the performance tests as specified in 3.3.2. Subsequent to the 1.5 hour vibration, the I. F. Amplifier shall be visually examined for physical damage, then tested as specified in 4.6.2.

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4.6.1.3 Humidity.- The humidity test shall be performed as specified in MIL-STD-810, Method 507, Procedure I for four (4) hours at the conditions specified in 3.3.3. Subsequent to this test, the I. F. Amplifier shall be examined for damage then tested as specified in 4.6.2.

4.6.1.4 Sand and Dust.- The sand and dust test shall be performed as specified in MIL-STD-810, Method 510, with the following conditions: The sand and dust particle sizes, density, wind velocity, temperature, time duration and relative humidity shall be as specified in 3.3.4. Subsequent to this test, the I. F. Amplifier shall be visually examined for damage then tested as specified in 4.6.2.

4.6.1.5 Altitude-Operating.- The altitude test shall be performed as specified in MIL-F-13926, Procedure I at an altitude of 10,000 feet and as specified in 3.3.5.1. The I. F. Amplifier shall be exposed to the above condition for a time period required to allow all entrapped air in the chamber to escape as evidenced by chamber pressure stabilization. The barometric pressure shall then be maintained to simulate 10,000 feet (20.58 inches of mercury) altitude condition. While the I. F. Amplifier is maintained at this pressure for four (4) hours, the I. F. Amplifier shall be subjected to the tests specified in 4.6.2.

4.6.5.1 Altitude-Non-operating.- The I. F. Amplifier shall be subjected to barometric pressure specified in MIL-F-13926, Procedure II for four (4) hours as specified in 3.3.5.2. Subsequent to this test, the I. F. Amplifier shall be visually examined for damage, then tested as specified in 4.6.2.

4.6.1.6 Fungus.- One (1) I. F. Amplifier shall be subjected to the fungus test specified in MIL-F-13926. At the end of the 28 day incubation period, the I. F. Amplifier shall be tested as specified in 4.6.2 for compliance with 3.3.6. Subsequent acceptance of I. F. Amplifier shall be by certification.

4.6.1.7 Temperature Cycling.

4.6.1.7.1 Storage Temperatures.- This test shall be performed as specified in MIL-F-13926, Procedure II, and thermally stabilized at +160°F for eight (8) hours and at -80°F for eight (8) hours. The I. F. Amplifier shall be returned to room ambient temperature +75 ±15°F, visually examined for damage then tested as specified in 4.6.2.

4.6.1.7.2 Operating Temperatures.- The I. F. Amplifier shall be placed in a test chamber and the temperature gradually raised until thermally stabilized at +140°F for eight (8) hours. The I. F. Amplifier shall be tested as specified in 4.6.2 during the first hour, at the four (4) hour period, and within the last hour of the eight (8) hour period. The temperature shall then be reduced to -50°F. When thermally stabilized at this temperature, the I. F. Amplifier shall be tested as specified in 4.6.2 at the three (3) time periods specified above. A minimum of five (5) to six (6) minutes warm-up period shall occur prior to performance of the first test. The I.F. Amplifier shall be returned to room ambient temperature +75 ± 15°F, examined for damage then tested as specified in 4.6.2.

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4.6.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily accomplished on every I. F. Amplifier at Standard Ambient Conditions. Standard Ambient Conditions for the purposes of these tests shall be defined as:

(a) Temperature between 60°F and 90°F.

(b) Barometric pressure between 28 and 32 inches of mercury.

(c) Relative humidity (RH) up to 90%.

4.6.2.1 Initial Conditions.- The following voltage and load connections shall be applied as diagrammed in Figure I.

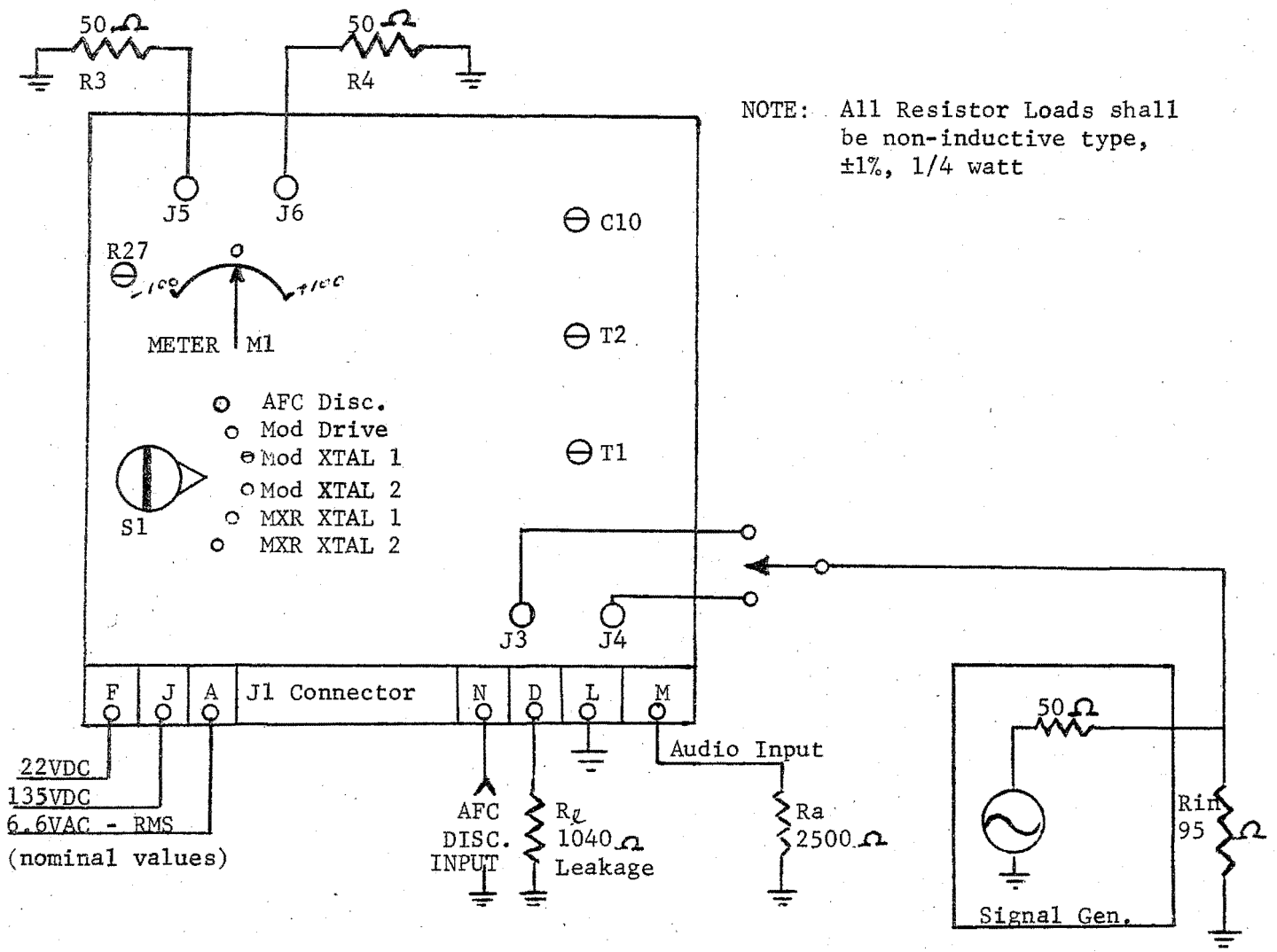


FIGURE I

4.6.2.2 Power Supply.- All requirements of paragraph 4.6.2 shall be met under the following power supply conditions except where an exception is specifically delineated.

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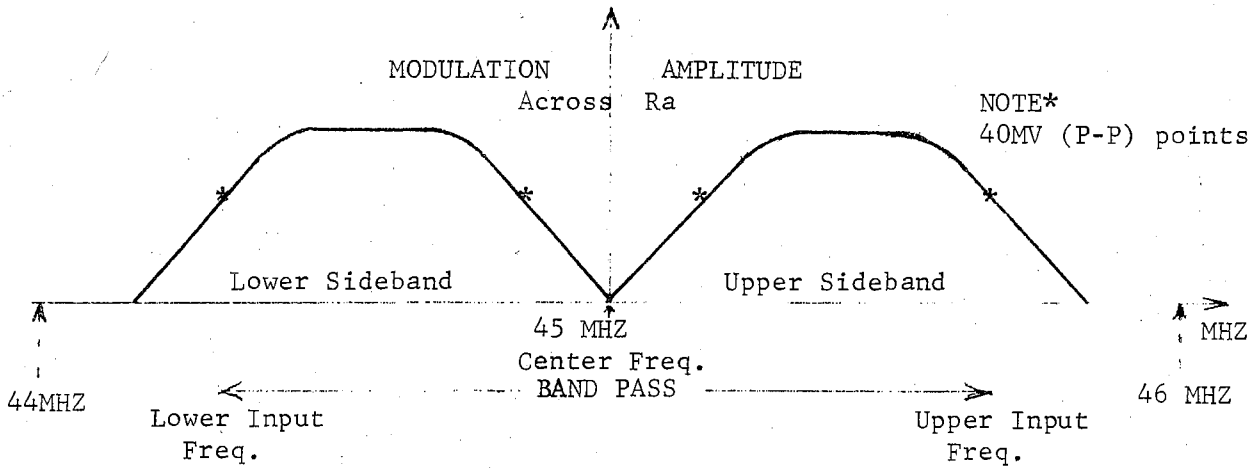
Nominal	+22.0 VDC	+135 VDC	6.6 VAC (RMS)
Condition I	+21.5 VDC	+130 VDC	6.3 VAC (RMS)
Condition 2	+22.5 VDC	+140 VDC	6.9 VAC (RMS)

4.6.2.3 Local Oscillator Frequency.- Adjust potentiometer R27 fully clockwise. Monitor the local oscillator frequency across external resistors R3 and R4. The local oscillator frequency shall be 45 megahertz plus or minus 0.01 percent after a five (5) to six (6) minute warm-up.

4.6.2.4 Modulator Drive.- Measure the voltage and frequency across R3 and R4 while varying adjustment potentiometer R27. The voltage shall be capable of being adjusted to 0.2 and 0.5 volts (RMS) with a frequency of 45 megahertz plus or minus 0.01 percent.

4.6.2.4.1 Modulator Drive Level.- Adjust potentiometer R27 to 0.4 volts (RMS) at a frequency of 45 megahertz plus or minus 0.01 percent with the nominal voltages of para 4.6.2.2. Monitor the above conditions for 20 minutes.

4.6.2.5 Bandpass and I. F. Alignment.- With a signal generator (50 ohm output impedance) and R-in connected as shown in Figure I, apply a 100 microvolt (RMS) 44.995,000 hertz signal to J3. Increase the 44.995,000 hertz input signal until the modulation frequency across R_a is 120 kilohertz. Adjust T1, T2 and C10 for a maximum amplitude. Vary the input signal between 44 and 46 megahertz and record the four (4) input frequencies where the amplitude across R_a is 40 millivolts peak-to-peak. The bandpass shall be between 0.26 and 1.0 megahertz with a center frequency of 45 megahertz ± 0.01 percent as diagrammed in Figure 2.



I. F. AMPLIFIER PEAKED AT MODULATION FREQUENCY OF 120 KILOHERTZ

FIGURE 2

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4 6 2 6 Gain.- Modulation Frequency Range of 10 to 120 kilohertz.

4.6.2.6.1 Signal applied to J3.- With a signal generator (50 ohm output impedance) and R-in connected as diagrammed in Figure I., apply a 100 microvolt (RMS) 44,995,000 hertz signal to J3. Monitor the local oscillator frequency across R3 to the requirements of paragraph 3.5.4. The difference frequency between the applied 44,995,000 hertz signal and the local oscillator frequency is the modulation frequency which should be present across Ra. Increase the frequency of the applied signal above 44,995,000 hertz so that the modulation frequency range of 10 to 120 kilohertz is present across Ra. The modulation frequency shall be an undistorted sine wave having an amplitude between 65 and 100 millivolts peak-to-peak over the modulation frequency range of 10 to 75 kilohertz and an amplitude between 55 and 100 millivolts peak-to-peak over the modulation frequency range of 75 to 120 kilohertz.

4.6.2.6.2 Signal applied to J4.- Repeat the tests of para. 4.6.2.6.1 except apply the input signal to J4 instead of J3 and monitor the local oscillator frequency across R4 instead of R3.

4.6.2.7 Noise and Distortion.- The output noise measured across Ra shall be less than 0.8 millivolts peak-to-peak. With a signal generator (50 ohm output impedance) and Rm connected as diagrammed in Figure I., vary the input level to J3 and then J4 between 5 and 500 microvolts (RMS) at a modulation frequency of 10, 75, and 120 kilohertz across Ra and insure that the output across Ra is an undistorted sine wave.

4.6.2.8 Leakage.- The voltage across R_1 (1040 ohms) with no input signal shall be between +67 and +73 millivolts DC.

4.6.2.9 AFC DISC Meter.- Place the Monitor Selector Switch (S1) in the AFC DISC position. With a +195 plus or minus 5 millivolt D. C. voltage applied to J1-N, the Monitor Meter (M1) shall indicate between +45 and +55 micro-amperes. With a -195 plus or minus 5 millivolt D. C. voltage applied to J1-N, the Monitor Meter (M1) shall indicate between -45 and -55 micro-amperes.

4.6.2.10 MOD DRIVE Meter.- Place the Monitor Selector Switch (S1) in the MOD DRIVE position. The Monitor Meter (M1) shall indicate between +40 and +70 micro-amperes after the accomplishment of 4.6.2.4.1.

4.6.2.11 MOD XTAL Meter.- Remove resistor loads R3 and R4 (50 ohms) from J5 and J6. As diagrammed in Figure 3, with the Monitor Selector Switch (S1) in the indicated position and a IN263 diode connected across J5 or J6 (observe polarity), the Monitor Meter (M1) shall read as indicated after the requirements of 4.6.2.4.1 have been accomplished.

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MONITOR SELECTOR SWITCH POSITION	DIODE CONNECTION	METER READINGS MICRO AMPS
MOD XTAL 1	1N263 	-50 to -100
MOD XTAL 1	1N263 	+50 to +100
MOD XTAL 2	1N263 	-50 to -100
MOD XTAL 2	1N263 	+50 to +100

FIGURE 3

4.6.2.12 MXR XTAL Meter.- Remove resistor load R_m (95 ohms) from the 45 megahertz signal generator. As diagrammed in Figure 4, apply a 320 millivolt (RMS) 45 megahertz plus or minus 0.01 percent signal to J4 or J3, connect a 1N263 diode (observe polarity) to J4 or J3, with the Monitor Selector Switch (S1) properly positioned, the Monitor Meter (M1) shall read as indicated.

SIGNAL GENERATOR CONNECTION	DIODE CONNECTION 1N263	MONITOR SELECTOR SWITCH	METER READINGS MICRO AMPS
J4	J4	MXR XTAL 1	-50 to -100
J4	J4	MXR XTAL 1	+50 to +100
J3	J3	MXR XTAL 2	-50 to -100
J3	J3	MXR XTAL 2	+50 to +100

FIGURE 4

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4.6.3 Electrical.

4.6.3.1 Continuity.- Continuity shall be tested by standard measuring equipment for compliance with 3.4.1.

4.6.4 Reliability Run-in.- The following Run-in tests shall be performed on a 100% basis.

4.6.4.1 Temperature Run-in.- The I. F. Amplifier shall be placed in a chamber and the temperature gradually raised until thermally stabilized at plus 125°F. After temperature stabilization, power shall be applied and maintained for three (3) hours. Five (5) minutes after power is applied, a complete test to the requirements of para 4.6.2 shall commence. A second test to the requirements of para 4.6.2 shall be accomplished during the second hour and a third test to the requirements of para 4.6.2 shall be accomplished during the third hour of the three (3) hour test. Repeat the above tests at a temperature of minus 25°F. Subsequent to the above temperature extremes (plus 125°F and minus 25°F) the I. F. Amplifier shall not have been damaged or adversely affected and shall meet the requirements of 4.6.2.

4.6.4.2 Vibration Run-in.- The vibration test shall be performed as specified in MIL-STD-810, Method 514 at the conditions specified in 3.6.2. During the above vibration test, the I. F. Amplifier shall meet the performance tests as specified in para 3.6.2. Subsequent to the vibration test, the I. F. Amplifier shall be visually examined for damage and shall not be damaged or adversely affected and shall meet the performance requirements of para 4.6.2.

4.6.4.3 Shock Run-in.- The shock test shall be performed as specified in MIL-STD-810, Method 516, Procedure IV at the conditions specified in 3.3.1. Subsequent to the eighteen (18) shocks, the I. F. Amplifier shall be visually examined for damage and shall not be damaged or adversely affected and shall meet the requirements of 4.6.2.

4.6.4.4 Failures.- Failures occurring during Reliability Run-in shall not be cause for rejection, but shall be corrected as outlined in para 3.6.4.

4.6.5 Packaging, Packing and Marking.- Quality Assurance Provisions for packaging, packing and marking shall be as specified in MIL-P-14232 and Section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet 10525674. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The I. F. Amplifier is used as a subassembly in the M36 Radar Chronograph.

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6.2 Ordering Data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Selection of applicable levels of preservation, packaging and packing
(c) Applicable stock number
(d) Applicable packaging data sheet (see 5.1)

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1290-A197

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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